Owen Wells

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https://owendwells.github.io/

Education

University of California Los Angeles
Mechanical Engineering, B.S. - GPA: 3.9 — Aug 2025 - Jun 2027
Mt. San Antonio College
Associate's of Science Transfer Mechanical Engineering & Mathematics - GPA: 4.0 — Jul 2022 - Jun 2025

Technical Skills

CAD: SolidWorks, Fusion, Autocad, Onshape, SolidWorks Assembly, SOLIDCAM
Analysis: Ansys Mechanical FEA, Structural Analysis, Simulink
Manufacturing: Lathe and Mill, Drill Press, GD&T, 3D printing/Rapid Prototyping, Heat Treatment, CNC Machining
Electronics: Arduino, Raspberry Pi, ESP32, Wiring Schematics, Soldering, Electronic Circuits, Sensor Integration
Programming: MATLAB, Python, C++, Microsoft Office, Excel

Experience

Bruin Formula Racing Steering Subsystem Lead, University of California Los Angeles — May 2026 - Present
- Designed steering column with single U-joint angle of 20° to minimize compliance and maintain sufficient driver ergo.
- Reduced steering system weight by .2 lbs, and streamlined a single U-joint steering column design for ease of assembly, serviceability and reliability at the cost of linear steering.
- Forecasted the benefits of anti-ackerman steering geometry with the vehicle dynamics lead to increase speed in corners by 3% higher lateral acceleration from the previous parallel design.

Bruin Formula Racing New Member FSAE, University of California Los Angeles — Sept 2025 - April 2026
- Created part drawings for the suspension subteam during manufacturing phase following ANSI standards and GD&T.
- Manufactured 30+ parts with manual machining and designed welding tab jigs in SOLIDWORKS for suspension.
- In charge of the assembly drawings and cost reports for the steering subsystem creating exploded view and BOM

Embedded Tutor / Drop In Tutor, Mt. SAC ASAC - Walnut, CA — Aug 2024 - Jun 2025
- Demonstrated exceptional level of knowledge of material by tutoring all levels of physics offered on campus, various engineering courses, and math up to calculus III.
- Inspired students by actively discovering their strengths to successfully guide them through the course.
- Led group tutoring sessions by facilitating different activities for students to remain engaged and collaborative.

Projects

Carbon Fiber A-Arms, Bruin Formula Racing — Oct 2025 – Present
- Designed carbon fiber A-arms for manufacturing, assembly and serviceability as part of a new member project with the goal of reducing weight by 70%. Tested alongside the current Steel design to compare strength and manufacturability.
- Optimized diameter and length geometry to minimize compressive yield margin of safety with a max force of 406 lbs.
- Designed and iterated aluminum wafers for the A-arms to cut more weight, using ANSYS mechanical FEA to identify stress fields and minimize weight and margin of safety, resulting in a final max force of 62216 psi on the load bearing.

Electronics & CAD Lead, Spy Watch, 2025 UCLA Engineering HACK Competition — Jul 2025 - Aug 2025
- Programmed a Raspberry Pi Pico to acquire sensor data, transmit to a web interface, and display feedback via LCD.
- Designed compact enclosure geometry to optimize component layout, sensor accessibility, and structural integrity.
- Designed and integrated a DC voltage regulator circuit to step 3.7V Lithium ion battery output to regulated 5V, ensuring reliable ESP32 CAM operation under Wifi current spikes.

Water Level Indicator & Regulator Breadboard Project — Oct 2024 - Dec 2024
- Designed and built a breadboard-based monitoring and control circuit to detect three discrete water levels with float switches, and three temperature ranges, driving LED indicators and an automated drain pump.
- Implemented a Wheatstone bridge with thermistor to normalize voltage at room temperature, followed by a difference amplifier (2 V/V) and comparator network to trigger temperature LEDs at defined thresholds.
- Designed a pump shutoff delay circuit using an RC time constant and NPN power transistor, enabling controlled drainage below the maximum fill level for 10 s, based on the calculated flow rate of the pump being 30 mL/s.